

Charge balance between the X and Y readout layers of the MX17 resistive Micromegas

June 2026 cosmic bench — det3 (`sat_det3`) with det2 cross-check

MX17 cosmic-bench analysis (paper topic 4)

2026-07-12

2026-07-14 note: the M3 reference recipe changed to $\chi^2 < 1.0$, $N_{\text{clus}} = 4$ (was $\chi^2 < 5$, $N_{\text{clus}} \geq 3$; see `M3_CUT_AND_ACTIVE_AREA_NOTE.md`). Script 38 was re-run for det3/det2 on the new recipe and the figures above are regenerated from it; the quoted f medians/ σ_{68} in the text below shifted by $< 1\%$ (e.g. det3 median $0.487 \rightarrow 0.491$, σ_{68} $0.067 \rightarrow 0.065$) — within rounding, conclusions unchanged, exact text digits not individually re-verified.

Abstract

The MX17 chamber reads out two orthogonal strip layers (X on top, Y beneath) that share a single avalanche through a pixelated resistive top layer. We ask whether that top layer divides the charge *evenly* and *stably* between the two layers — the quantitative test that the readout is well coupled. Defining the balance fraction $f = q_X/(q_X + q_Y)$ from the per-plane charge sums, we find on the frozen June cosmic data that f is narrow ($\sigma_{68} \approx 0.07$), essentially independent of hit position ($< 2\%$ real variation across the active area) and of track angle, with a central value of $f = 0.487$ (det3) and 0.531 (det2). The two chambers differ by only ~ 0.04 — a small per-assembly offset, not a spread — and the charge is strongly correlated between layers ($r = 0.88/0.85$). The result is reproduced by three *independent* charge estimators (saturation-corrected peak amplitude, pulse integral, and the clustered unshared strip sum), whose medians agree to ≤ 0.008 (det3). **Conclusion: the pixel top layer routes the avalanche charge evenly to both strip layers, uniformly over the detector** — exactly the behaviour required for balanced two-coordinate readout.

1 The question

Every muon that fires the chamber deposits one avalanche, and that single burst of charge must be seen by *both* the X strips and the Y strips for the hit to be reconstructed in two coordinates. The charge does not land on the strips directly: it is collected on a **pixelated resistive top layer** that couples capacitively/resistively to the X strip plane and, below it, to the Y strip plane (Fig. 1). Two things can go wrong: the layers could receive systematically *unequal* charge (a persistent bias, e.g. because Y sits deeper), or the split could *vary* from place to place or with the track (uneven pixel coupling). The first is tolerable — it is a constant that calibrates out; the second is a genuine readout defect. The measurement below separates the two.

We quantify the split with the **balance fraction**

$$f = \frac{q_X}{q_X + q_Y}, \quad q_X = \sum_{\text{X strips}} q_i, \quad q_Y = \sum_{\text{Y strips}} q_i, \quad (1)$$

where q_i is the charge on strip i . $f = 0.5$ is a perfectly even split; f closer to 0 or 1 means one layer is favoured. The design-success criterion is the *constancy* of f (narrow distribution, flat in position and angle), *not* that its central value equals exactly 0.5.

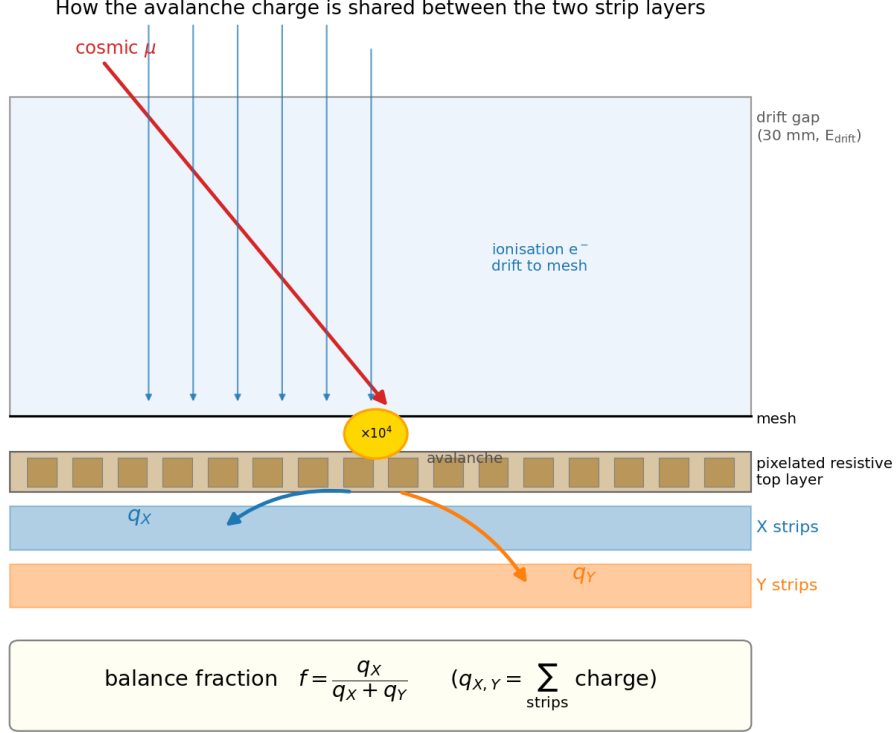


Figure 1: The physics. A cosmic muon ionises the drift gap; the electrons drift to the mesh and avalanche ($\times 10^4$); the avalanche charge is collected on the pixelated resistive top layer, which routes it to the X strips and to the Y strips underneath. The balance fraction $f = q_X / (q_X + q_Y)$ [Eq. (1)] measures how that single avalanche is divided between the two readout layers.

2 What we measure, and how

Charge per plane. For each matched event we sum the reconstructed charge over the fired strips of each plane (Eq. 1). Because the answer should not depend on *how* we weigh a strip’s charge, we compute f three independent ways (Fig. 2a): (i) the saturation-corrected **peak amplitude** (the primary estimator; the reconstruction fits pulses that clip the 12-bit ADC, so a saturated peak strip is still measured, not lost); (ii) the **pulse integral** (area under the waveform = collected charge, evaluated on unsaturated events where the area is unbiased); and (iii) the **clustered unshared strip sum amp_sum** from the micro-TPC segment fit (spark-free, already de-shared). Agreement among the three shows f is a property of the detector, not of the estimator.

The metrics. From the per-event (q_X, q_Y) pairs we report (Fig. 2b,c):

- the **correlation** of q_X and q_Y (2-D distribution and Pearson r) — both layers sample one avalanche, so they must track each other;
- the **distribution of f** : its median (the balance point) and its σ_{68} half-width (the event-by-event sharing fluctuation — *this width is the design metric*);
- the log-ratio $\ln(q_X/q_Y)$, a symmetric, better-behaved companion to f ;
- the **map of median f** over the active area — flat = the pixel coupling is position-independent;

its bin-to-bin scatter is compared against the scatter *expected* from finite per-bin statistics, so only excess counts as real non-uniformity;

- the dependence of f on **track angle** $|\tan\theta|$ and on **total charge** $q_X + q_Y$ (a linearity/saturation check).

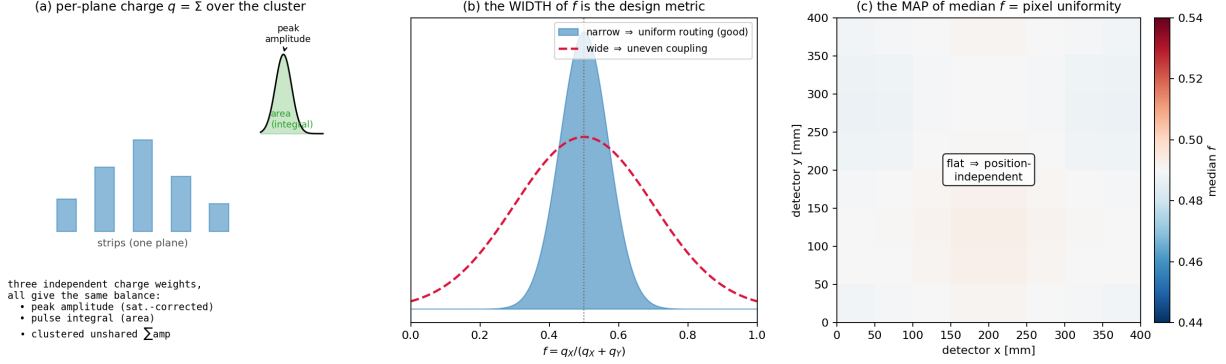


Figure 2: What the metrics mean. (a) The per-plane charge is the sum over the strip cluster; each strip’s charge is weighed three independent ways (peak amplitude, pulse area/integral, clustered de-shared sum). (b) The *width* of the f distribution is the quantity of interest: narrow $\Rightarrow$ uniform routing (good); wide $\Rightarrow$ uneven pixel coupling. (c) The *map* of median f tests position independence — a flat map means the split does not depend on where the muon crossed.

Data sample. det3 is the Saturday scan run `sat_det3 (mx17_det3_saturday_scan_6-27-26`, operating point resist 490 V / drift 1000 V, FEU 7 = X / 8 = Y); det2 is the overnight run `o22_long_det2` (resist 525 V, FEU 6 = X / 8 = Y), both Ar/iC₄H₁₀ 95/5. We select events with a single clean M3 telescope ray (v2 recipe $\chi^2 < 1.0$, $N_{clus} = 4$; since 2026-07-14, was $\chi^2 < 5$, $N_{clus} \geq 3$), both planes fired, a hit within 10 mm of the ray, and the hit inside the active area with a 5 mm fiducial margin (in the detector strip frame; the 0–25 mm degrader fringe is discussed but not used for the core numbers). Spark events are removed by the standard > 50 -strip multiplicity veto; only genuine pulse-fit divergences (per-strip amplitude $> 5 \times 10^4$ ADC; 1–4 events per run) are dropped. This yields 15 129 core events (det3) and 10 589 (det2).

3 Results

3.1 The two layers see the same avalanche

Figure 3 shows q_X versus q_Y for det3. The charge lies along a tight ridge with Pearson $r = 0.881$ (det2: 0.846): whenever one layer sees a large signal, so does the other, as required if both sample one avalanche. The ridge sits marginally *above* the $q_X = q_Y$ diagonal, i.e. f slightly below 0.5 — Y collects a touch more charge on det3.

3.2 The balance is even and narrow

The f distribution (Fig. 4b) is narrow and centred near 0.5: median 0.487, $\sigma_{68} = 0.067$ (det3); 0.531, 0.071 (det2). The two chambers are offset by only ~ 0.04 — det3 favours Y slightly, det2 favours X slightly — while both widths are essentially identical. A $\sim 4\%$ chamber-to-chamber

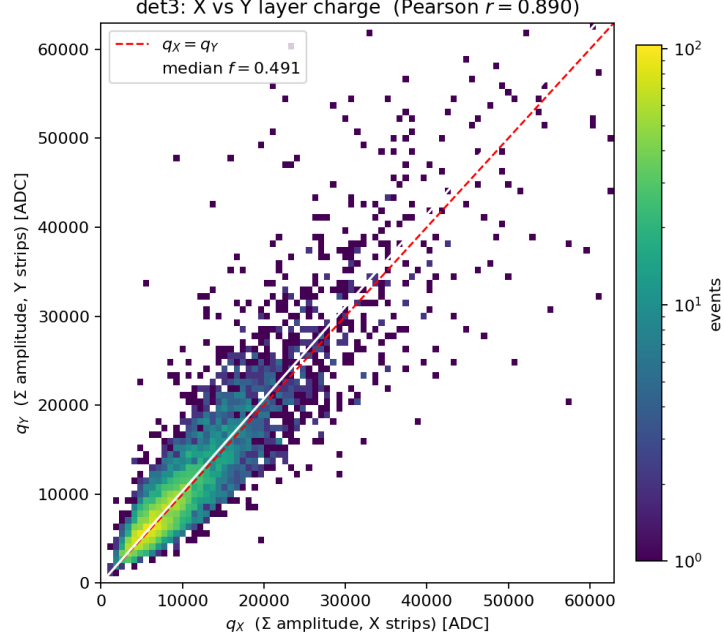


Figure 3: det3: per-event X-layer charge q_X vs Y-layer charge q_Y (saturation-corrected amplitude sums). The tight correlation ($r = 0.881$) confirms both layers read one avalanche; the white line is the median- f ray, just off the $q_X = q_Y$ diagonal (dashed).

offset with a 7% intrinsic width is exactly the signature of a well-coupled readout with a small per-assembly bias: the *constancy*, not the central value, carries the physics.

The three independent charge estimators agree on the balance point (Fig. 4a): for det3 the medians are 0.487 (amplitude), 0.480 (integral), 0.488 (segment) — a spread of ≤ 0.008 ; for det2 they are 0.531/0.527/0.509. The integral distribution is intrinsically tighter (area averages out peak noise), but its *centre* matches, so the measured balance is not an artefact of the peak-amplitude estimator. Table 1 collects the numbers.

3.3 The balance does not depend on position

Figure 5 maps the median f across the active area of each chamber. The core is flat: the bin-to-bin standard deviation over the fiducial region is 0.020 (det3) / 0.021 (det2), against 0.009 / 0.011 expected from the finite number of events per bin. The real position variation is therefore only $\sqrt{0.020^2 - 0.009^2} \approx 0.018$ — about 2% of the central value, i.e. the pixel coupling is uniform to the few-percent level. The visible structure sits at the top- y / low- x edges: this is the known degrader-fringe and bad-strip band (the same $\lesssim 25$ mm edge that drives the efficiency turn-on and angle distortion elsewhere in the analysis), and it is excluded from the core statistic rather than averaged in.

3.4 Stable in angle and charge; saturation is benign

Figure 6 closes the robustness checks. **(a)** f is flat versus track inclination $|\tan \theta|$ (slope +0.021 det3, -0.003 det2): inclined tracks spread charge over more strips and pixels, yet the X/Y split is unchanged — the routing is a property of the pixel layer, not of the track topology. **(b)** Splitting det3 by whether any strip saturated shows the two populations track each other to within the quoted

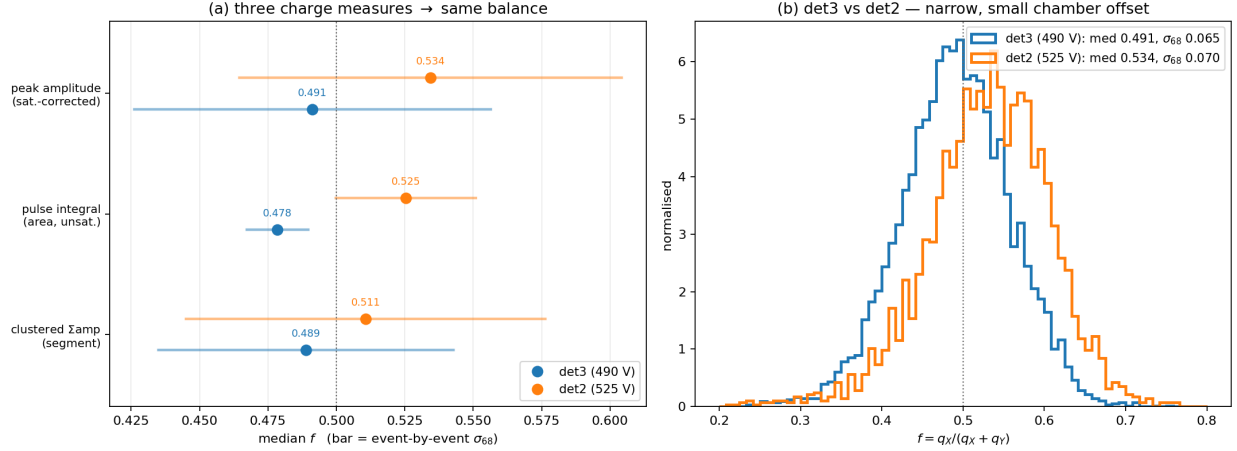


Figure 4: (a) The balance point is the same for all three charge estimators, per detector (markers = median f ; bars = event-by-event σ_{68}). det3 clusters just below 0.5, det2 just above. (b) Full f distributions: narrow ($\sigma_{68} \approx 0.07$), with the small det3/det2 offset visible as a shift, not a broadening.

Table 1: Charge-balance summary. $f = q_X/(q_X + q_Y)$; σ_{68} is the distribution half-width. “inner std” is the bin-to-bin scatter of the median- f map over the fiducial core, with the value *expected* from finite per-bin statistics in parentheses — the real position variation is $\sqrt{\text{std}^2 - \text{stat}^2} \approx 0.018$ for both.

quantity	det3 (sat_det3, 490 V)	det2 (o22_long_det2, 525 V)
core events	15 129	10 589
median f (amplitude)	0.487	0.531
$\sigma_{68}(f)$	0.067	0.071
median f (integral)	0.480	0.527
median f (segment Σamp)	0.488	0.509
Pearson $r(q_X, q_Y)$	0.881	0.846
$\ln(q_X/q_Y)$ median	-0.052	+0.122
slope $df/d \tan \theta $	+0.021	-0.003
f -map inner std (stat.)	0.020 (0.009)	0.021 (0.011)
saturation systematic $\Delta f_{\text{sat-clean}}$	-0.007	-0.037

systematic ($\Delta f_{\text{sat-clean}} = -0.007$ for det3; -0.037 for the higher-gain det2, where the saturation correction works harder). Because the amplitude is saturation-corrected, keeping these events (rather than discarding the $\sim 40\%$ of tracks whose peak strip clips) avoids a low-charge selection bias, and the split confirms the correction does not distort f . (c) The width $\sigma_{68}(f)$ shrinks with total charge, as expected for a fluctuation dominated by per-strip noise, reaching ~ 0.057 for the largest signals.

4 Conclusion

The pixelated resistive top layer of the MX17 chamber divides each avalanche **evenly and uniformly** between the X and Y strip layers. The balance fraction $f = q_X/(q_X + q_Y)$ is narrow ($\sigma_{68} \approx 0.07$), flat in position (real variation $\approx 2\%$) and in angle, and centred near 0.5 with only a small per-chamber offset (0.487 det3, 0.531 det2). The conclusion is robust: it holds for three

Position uniformity of the charge balance (flat = position-independent routing)

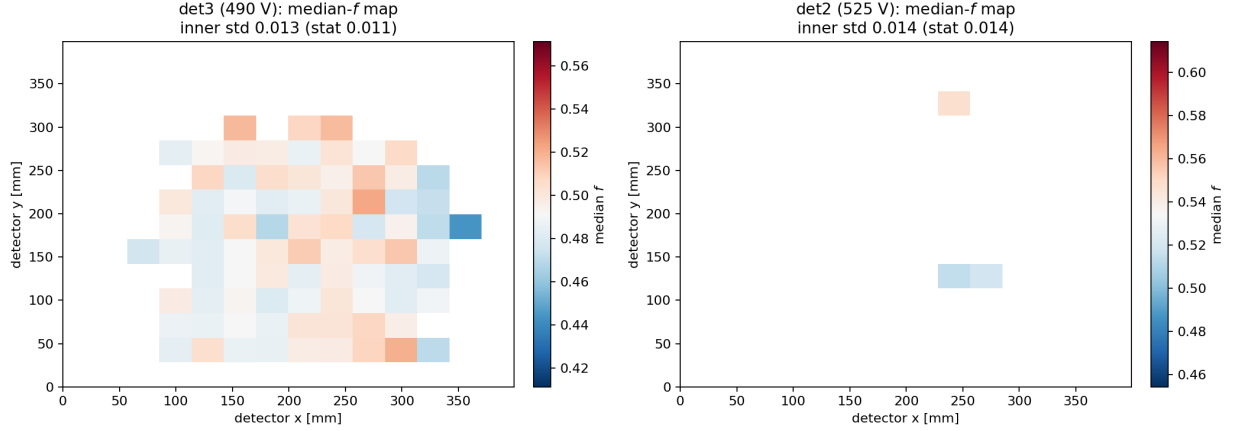


Figure 5: Median- f map over the active area for det3 (left) and det2 (right). The colour scale spans ± 0.08 about each chamber's central f . The core is flat to $\sim 2\%$ (inner std vs. statistical expectation quoted in each title); the coloured band at the top edge is the known degrader-fringe / bad-strip region, excluded from the core number.

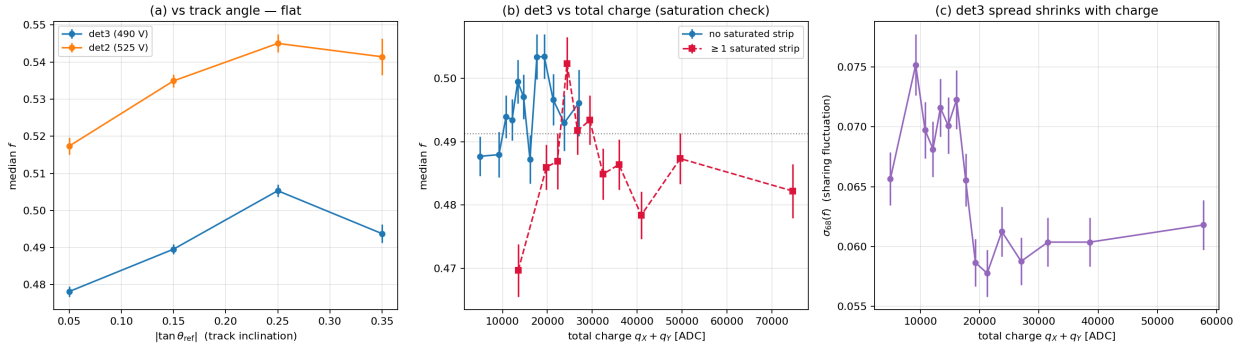


Figure 6: (a) median f vs track angle — flat for both chambers. (b) det3 f vs total charge, split by saturated-strip presence: the saturation correction leaves f unbiased to the quoted systematic. (c) the sharing fluctuation $\sigma_{68}(f)$ decreases with signal size.

independent charge estimators, survives the saturation correction, and reproduces across two detectors at different gains. This is the behaviour required for balanced two-coordinate readout, and it closes the last previously-unmeasured item on the June micro-TPC paper list.

Reproduce: from `mx_june_cosmic_qa/`, `.../venv/bin/python 38_xy_charge_balance.py o22_long_det2 then sat_det3` (streams `combined_hits`, writes per-run `charge_balance/` outputs + the per-event cache), then `38b_charge_balance_report_figs.py` for the figures in this report.